LOT Pager ??? 01-PLAN-ROADMAP

01 · Plan & Roadmap — LOT Pager

1. Concept

A 40×40 mm stainless-steel “pager” for lot-systems.com. The site’s AI decides when to nudge the user (“Coffee time!”); the device shows it on a small screen. Pressing the button acts as **Copy** and writes a `device_copy` entry into the user’s **Log** tab. Camera + weather + motion sensors feed back context (opt-in).

2. Requirements trace (your 19 points → where handled)

#	Requirement	Where
1	PCBWay	06 (PCB, SMT, CNC)
2	Pager-like AI notifications	04, 05
3	2-part stainless body	03
4	4×4 cm × 5 mm silver flat square	03 (thickness analysis — see risk R1)
5	Camera	02, 03, 04
6	LOT API connector	05
7	PDF manuals	`pdf/` build (README)
8	Compress info each session	`STATE.md` + digest at top of each report
9	Firmware documents	04
10	Software to connect with firmware	04 §7 (LOT Device Tool)
11	Separate documents	one file per topic
12, 19	Charger / wireless charger	02, 03 (R2)
13	100-unit run	06
14, 15	Weather + AI-grade off-the-shelf sensors	02
16	Button “Copy” → Log tab	05
17, 18	Polished back; camera+screen+button front	03

3. Roadmap

Phase	Weeks	Output	Exit gate
P0 Plan	0	This package	Vadik approves BOM tier A
P1 Server	1–3	`/api/device/*`, `device_copy` log, pairing UI	Simulated device (script) round-trips in Log tab

Phase	Weeks	Output	Exit gate
P2 Dev-kit prototype	2–5	ESP32-S3 dev board + 1.54” TFT + OV2640 + button on breadboard; firmware v0.1	“Coffee time!” displays; Copy appears in Log
P3 EVT PCB	5–10	5–10 rigid PCBs (PCBWay), 3D-printed + CNC aluminium test shells	Fits 40×40; Qi + RF window test pass/fail known
P4 DVT	10–16	20 units, real 316L CNC bodies, polish, battery, certification pre-scan	Thickness, RF range, Qi ≥ 2.5 W at coil, 7-day battery
P5 PVT / pilot	16–22	100-unit run (06)	≥ 95 % units pass functional test
P6 Certify & ship	18–28	FCC/CE/UN38.3, manuals PDF, OTA	Certificates in hand

Critical path: **Qi/RF window decision (R2)** → steel body drawing freeze → 100-unit order. Do not order steel tooling before P3 results.

4. Risk register

ID	Risk	Likelihood	Impact	Mitigation
R1	5 mm total thickness infeasible	High	High	EVT at 7.5 mm; DVT tries 6 mm with bare-die ESP32-S3 + OV2640 CSP + 3 mm cell; keep 5 mm for the no-camera variant
R2	Steel shields Qi and 2.4 GHz	High	High	Dielectric (ceramic/glass/PEEK) windows: RF slot at the seam between the two halves, Qi window in back mirror (sapphire/ceramic disc) or contact-charge fallback
R3	Battery safety in sealed metal case	Med	High	Protected cell, BQ-class charger w/ NTC, UN38.3 certified cell, vent-free design review
R4	Camera privacy/regulation	Med	Med	Hardware indicator LED tied to camera power, default OFF, explicit opt-in in site settings
R5	Polished mirror shows	High	Low	#8 mirror finish + optional PVD

ID	Risk	Likelihood	Impact	Mitigation
	fingerprints/scratches			coating; document in manual
R6	Unit cost > target	Med	Med	See 02 totals; steel is ~40 % — evaluate MIM or stamped back plate at volume
R7	Brand assets not yet read	—	Low	HW-02 task

5. Success metrics

- Notification latency (site → screen): < 5 s while on Wi-Fi, < 40 s in power-save.
- Copy press → Log entry visible: < 3 s.
- Battery: ≥ 5 days typical (screen wakes ≤ 30×/day, 30-min sensor cadence).
- 100-unit yield ≥ 95 %.